The Norm Blind Spot of Amplitude-Encoding Quantum Kernels in Anomaly Detection

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Abstract

Amplitude encoding is a common way to load classical data into quantum kernel methods. We show that any kernel machine built on it scores an input by a Rayleigh quotient of a learned matrix. The score depends on the direction of the input and never on its norm. We ask when this bias helps anomaly detection. We test it on four security datasets from QKD telemetry and network flows, with every model trained on the same data at every sample size. In distribution the amplitude kernel never beats a Gaussian kernel, and it needs more data when samples are few. On attack types held out from training the outcome depends on the data. In both simulated settings the kernel fails, because unseen attacks differ from normal traffic mainly in norm. Appending one constant amplitude, set to the median training norm, repairs the failure. We explain why this constant works. Where the norm carries no signal, the anchor can cost accuracy. On UNSW-NB15 and CIC-IDS2017 the plain kernel does not fail. An eight-qubit ZZ feature map with tuned bandwidth is competitive with tuned classical kernels but is not the best on held-out attacks in any of six settings. Estimating every kernel entry from 1000 shots lowers mean AUC by at most 0.015 in any setting. A simple norm statistic does not predict when the failure occurs. Quantum kernels should be checked for their invariances and tested on held-out classes from real data, against classical kernels trained at the same scale.

Keywords: Quantum kernels, Amplitude encoding, Inductive bias, Anomaly detection, Quantum key distribution, Benchmarking

1 Introduction

Quantum kernel methods map classical inputs to quantum states and use the overlap of two states as a kernel [1, 2]. Many quantum models for supervised learning can be written this way [2]. So the question of whether a quantum model helps often becomes a question about its kernel. Theory says the answer depends on how the kernel's geometry matches the data [3, 4]. Proven advantages exist only for problems built for the purpose [5]. Benchmarks on standard data rarely find them [6].

Security is one of the domains where quantum kernels are now being tried, for intrusion and botnet detection [7, 8]. Such studies face two problems. Circuit simulation is slow, so quantum models are often trained on a few dozen samples while classical baselines see thousands. Tests also rarely hold out attack types, although a deployed detector will meet attacks it never saw [9].

This paper studies amplitude encoding, the encoding most often used in that work. Its kernel can be computed exactly and cheaply on a classical computer. That lets us remove the first problem. We train it at every sample size on the same data as classical kernels. We then ask a question that does not need quantum hardware. Is the inductive bias of this kernel useful for detecting attacks, and when?

The answer turns on one property. An amplitude-encoded state forgets the norm of its input. We make this exact. Every kernel machine on this kernel scores an input by a Rayleigh quotient of a learned symmetric matrix (Proposition 2). The score is blind to the input's norm and to its sign. Anomaly detectors often rely on exactly this information, since unusual events tend to lie far from typical data.

We test the consequences in two domains. The first is telemetry from quantum key distribution (QKD), where implementation attacks [10–13] shift channel statistics. The second is network flow monitoring, on a simulator and on the public UNSW-NB15 and CIC-IDS2017 datasets [14, 15].

Our contributions are as follows.

1. We show that amplitude-encoding kernel machines compute Rayleigh quotients. This gives their exact invariances and their function class.
2. We compare the kernel with classical kernels and other learners at matched scale on four datasets.
3. We find that it fails on held-out attack types in both simulated settings, and we trace the failure to norm invariance through targeted interventions.
4. We propose a norm anchor that costs one amplitude and at most one qubit. We show why the median training norm is a good anchor.
5. We find that the failure does not occur on two public benchmarks, and that a simple norm statistic cannot tell the cases apart.
6. We compare an entangling ZZ feature map with tuned bandwidth, and we measure the effect of finite-shot kernel estimation.

Section 2 reviews related work. Section 3 derives the properties of the kernel and the anchor. Section 4 describes the data and protocol. Section 5 gives results and Section 6 discusses them.

2 Related work

Quantum kernels and their bias. Havlíček et al. proposed quantum feature maps whose kernels may be hard to compute classically [1]. Schuld showed that many supervised quantum models are kernel methods [2]. Huang et al. showed that data can let classical learners match quantum ones, and gave a geometric test for potential advantage [3]. Kübler et al. argued that quantum kernels need a carefully chosen inductive bias to generalize, since their feature spaces are very large [4]. The encoding largely sets that bias [16, 17]. Our analysis is an instance of this view, worked out exactly for amplitude encoding.

Concentration and bandwidth. Fidelity kernels on many qubits tend to concentrate. Off-diagonal entries shrink toward a constant, and exponentially many shots are then needed to resolve them [18]. Scaling the inputs by a bandwidth avoids this and restores generalization [19, 20]. Shot noise also limits what a kernel can learn on near-term hardware [21, 22]. We tune the bandwidth of our ZZ baseline and simulate finite shots for both quantum kernels.

Benchmarking and classical simulation. Bowles et al. found that out-of-the-box classical models usually beat quantum models on standard tasks [6]. Tang showed that some proposed quantum speed-ups vanish against classical algorithms with similar data access [23]. Amplitude encoding is classically easy in a stronger sense. Its kernel has a closed form (Proposition 1). Any benefit must come from its inductive bias, which is what we measure.

Machine learning for QKD and network security. QKD attacks are countered by hardware monitoring [24] and by protocols such as decoy states [25] and measurement-device-independent QKD [26]. Mao et al. detect attacks on continuous-variable QKD with neural networks [27]. Machine-learning intrusion detection is surveyed by Buczak and Guven [28]. Kitsune popularized autoencoders trained on benign traffic only [29]. Quantum kernels have been applied to intrusion detection [7, 8] without the matched-scale and held-out-attack tests we use here.

3 The amplitude-encoding kernel

3.1 Exact form and cost

Amplitude encoding pads $\mathbf{x} \in \mathbb{R}^d$ with zeros to 2^q entries, with $q = \lceil \log_2 d \rceil$. It normalizes the result into the q-qubit state $|\psi(\mathbf{x})\rangle = \sum_i \hat{x}_i |i\rangle$, where $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$.

Proposition 1 *The fidelity kernel of amplitude encoding is*

$$K_A(\mathbf{x}, \mathbf{y}) = |\langle \psi(\mathbf{x}) | \psi(\mathbf{y}) \rangle|^2 = (\hat{\mathbf{x}}^\mathsf{T} \hat{\mathbf{y}})^2. \tag{1}$$

It costs $\mathcal{O}(d)$ classical operations per pair.

Proof The amplitudes are real and the padding is zero, so the overlap equals $\hat{\mathbf{x}}^\mathsf{T} \hat{\mathbf{y}}$. □

On hardware, each entry would be estimated by a compute-uncompute or swap-test circuit. Preparing an arbitrary amplitude-encoded state needs $\mathcal{O}(2^q)$ two-qubit

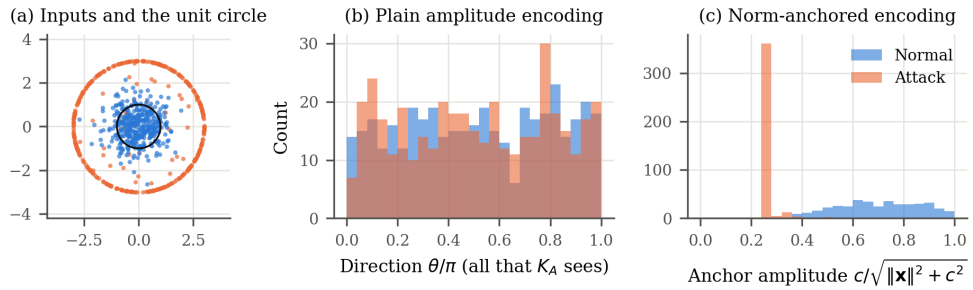


Fig. 1 Toy data in $\mathbb{R}^2$. Attacks are normal points pushed outward along the same directions. (a) Inputs and the unit circle onto which amplitude encoding projects them. (b) The direction, the only quantity K_A uses, does not separate the classes. (c) The extra amplitude added by the norm anchor separates them.

gates [30]. Estimating one entry to additive error ϵ needs $\mathcal{O}(1/\epsilon^2)$ shots. A training Gram matrix on n samples needs $n(n-1)/2$ such estimates. The closed form has none of these costs. We checked Eq. (1) against a PennyLane simulation [31] of the overlap circuit. The two agree to 7×10^{-15}. The simulated circuit takes 26 ms per entry and the closed form 3×10^{-4} ms.

3.2 The function class

Proposition 2 *Let $f(\mathbf{x}) = \sum_i a_i K_A(\mathbf{x}_i, \mathbf{x}) + b$ be any kernel expansion on K_A, as produced by an SVM, kernel ridge regression or a Gaussian process mean. Then*

$$f(\mathbf{x}) = \hat{\mathbf{x}}^{\mathsf{T}} W \hat{\mathbf{x}} + b, \qquad W = \sum_i a_i \, \hat{\mathbf{x}}_i \hat{\mathbf{x}}_i^{\mathsf{T}}. \tag{2}$$

So $f - b$ is the Rayleigh quotient of the symmetric matrix W at $\mathbf{x}$.

Proof $(\hat{\mathbf{x}}_i^{\mathsf{T}} \hat{\mathbf{x}})^2 = \hat{\mathbf{x}}^{\mathsf{T}} \hat{\mathbf{x}}_i \hat{\mathbf{x}}_i^{\mathsf{T}} \hat{\mathbf{x}}$. Sum over i. □

Corollary 1 *Such a model gives $\mathbf{x}$, $t\mathbf{x}$ and $-t\mathbf{x}$ the same score for every $t > 0$. Its scores lie between $\lambda_{\min}(W) + b$ and $\lambda_{\max}(W) + b$. Its feature space has dimension $d(d+1)/2$, the space of symmetric matrices.*

Proposition 2 holds for any training method and any regularization. We confirmed it numerically for a trained SVM to 10^{-14}. On standardized inputs, the origin is the training mean. The norm $\|\mathbf{x}\|$ is then the distance from typical data. The kernel cannot use it. The corollary makes a testable prediction. K_A should help when labels depend on direction alone. It should fail when an attack differs from normal traffic mainly in size. Figure 1 illustrates the second case.

3.3 The norm anchor

We append a constant before encoding,

$$\phi_c(\mathbf{x}) = \frac{(\mathbf{x},\, c)}{\sqrt{\|\mathbf{x}\|^2 + c^2}}, \qquad c = \text{median}_{\text{train}}\, \|\mathbf{x}\|. \tag{3}$$

The anchored kernel is

$$K_c(\mathbf{x}, \mathbf{y}) = \frac{(\mathbf{x}^\mathsf{T}\mathbf{y} + c^2)^2}{(\|\mathbf{x}\|^2 + c^2)(\|\mathbf{y}\|^2 + c^2)}, \tag{4}$$

a cosine-normalized inhomogeneous quadratic kernel. It needs one more amplitude. That adds a qubit only when d is a power of two.

Proposition 3 *Any kernel expansion on K_c has the form*

$$f(\mathbf{x}) = \frac{\mathbf{x}^\mathsf{T} A\mathbf{x} + 2c\,\mathbf{v}^\mathsf{T}\mathbf{x} + c^2 w}{\|\mathbf{x}\|^2 + c^2} + b \tag{5}$$

for a symmetric A, a vector $\mathbf{v}$ and a scalar w. Along a ray $\mathbf{x} = t\mathbf{u}$ with $\|\mathbf{u}\| = 1$, f moves from $w + b$ at $t = 0$ to $\mathbf{u}^\mathsf{T} A\mathbf{u} + b$ as $t \to \infty$.

Proof Apply Proposition 2 to $\phi_c(\mathbf{x})$ and split W into blocks A, $\mathbf{v}$ and w. □

The anchored model can therefore score by norm. With $A = I$, $\mathbf{v} = 0$ and $w = 0$ it computes $t^2/(t^2 + c^2)$, which rises with distance from the training mean. That curve reaches half its range at $t = c$, and its slope peaks at $t = c/\sqrt{3}$. The model's sensitivity to norm is concentrated around the scale c. If c is much smaller than typical norms, every input looks far away and norm is invisible again. If c is much larger, every input looks close to the origin and all states crowd together. The median training norm puts the steep region where normal data lies, so departures from it are visible. It uses training data only.

3.4 Sensitivity

The Jacobian of $\mathbf{x} \mapsto \hat{\mathbf{x}}$ is $(I - \hat{\mathbf{x}}\hat{\mathbf{x}}^\mathsf{T})/\|\mathbf{x}\|$. Its spectral norm is exactly $1/\|\mathbf{x}\|$. Normalization shrinks perturbations of large inputs and magnifies those near the origin. So amplitude encoding gives no uniform robustness guarantee [17].

4 Data and protocol

Testbeds. Table 1 lists the four datasets. Each QKD sample is one key-generation block of a decoy-state BB84 link over 50 km of fiber [25]. Normal operation drifts in loss, detector efficiency, misalignment, dark counts and background light. Error-rate estimates carry finite-size noise. Four attacks perturb seven measurements, each scaled by a strength s. We report the hard setting $s = 0.3$. Features are the seven

Table 1 Datasets. Public data uses 20,000 training and 50,000 test flows per seed in the main experiments.

Dataset	Samples	Features	Qubits	Attack types	Seeds
QKD telemetry	2,000	8	3	4	10
Synthetic flows	8,000	64	6	5	10
UNSW-NB15	257,673	68	7	9	5
CIC-IDS2017	2,520,730	69	7	6	5

measurements plus a robust Mahalanobis distance to normal blocks [32]. The flow simulator draws NetFlow-like marginals coupled by a Gaussian copula. Five attacks each shift 6 to 7 latent dimensions. UNSW-NB15 uses its official split. CIC-IDS2017 is cleaned of missing values, infinite values and 308,381 duplicates. Classes under 100 flows are dropped and the rest form six families. Public features pass through $\operatorname{sign}(x)\log(1+|x|)$. All inputs are standardized on training data.

Models. AE-SVM uses K_A. AE-SVM (anchored) uses K_c. AE-SVM (nonneg) scales inputs to $[0,1]$ before encoding. This removes the sign ambiguity but keeps norm invariance. Classical kernels are RBF, Laplacian and cubic polynomial, all in the same SVM. Above 1,500 samples we use a Nyström approximation [33]. Other models are random forests (RF) [34], XGBoost [35] and an MLP. On public data we add a deep network (DNN) and a benign-only autoencoder [29]. On QKD data we add a Laplace Gaussian process (GP) [36], isotonic RF [37] and three novelty detectors.

Entangling feature map. To test whether the findings are specific to amplitude encoding, we add the ZZ feature map of Havlíček et al. [1]. It applies $(U(\mathbf{z})H^{\otimes q})^2$ to $|0\rangle^{\otimes q}$, with

$$U(\mathbf{z}) = \exp\Big(i\sum_j \beta z_j Z_j + i\sum_j (\pi-\beta z_j)(\pi-\beta z_{j+1}) Z_j Z_{j+1}\Big). \tag{6}$$

We use $q = 8$ qubits with linear entanglement. Inputs with more than eight features are reduced by PCA fit on training data, then standardized. The bandwidth β is chosen by three-fold cross-validated AUC on the training data, which contains known attack types only. The candidates are 0.01, 0.02, 0.05, 0.1, 0.2, 0.4 and 0.8. The widths of an RBF kernel and a Laplacian kernel $\exp(-\gamma\|\mathbf{z}-\mathbf{z}'\|_1)$ are chosen the same way, with γq set to 0.1, 0.3, 1, 3 or 10. In this experiment all five kernels see the same eight inputs. Each SVM is trained exactly on 1,000 samples. QKD inputs are used as the QKD pipeline produces them, with the Mahalanobis distance non-negative. A control setting re-standardizes that column. We simulate the ZZ states exactly. They match PennyLane to 10^{-15}.

Finite shots. We model a compute-uncompute estimate of each kernel entry. An entry with true value K is replaced by B/S, where $B \sim \text{Binomial}(S, K)$ and S is the number of shots. Training Gram matrices are kept symmetric with unit diagonal. Test entries are sampled independently.

Protocol. Every model in a comparison is trained on the same subsample at every size. Zero-day tests leave one attack type out of training and test on normal traffic plus that type. We report ROC-AUC there, since it needs no threshold tuned on known attacks. Intervals are 95% confidence intervals over seeds. Code and data preparation are public (see Declarations).

5 Results

5.1 In-distribution detection

AE-SVM never beats RBF-SVM in distribution (Table 2). On QKD data their MCCs are 0.814 and 0.810, and McNemar's test finds no significant seed. On synthetic flows their accuracies are 0.947 and 0.948 ($p = 0.54$). On UNSW-NB15 they are 0.862 and 0.863 ($p = 0.24$). On CIC-IDS2017 RBF-SVM leads by 0.001 ($p = 0.001$). Tree ensembles are strongest overall. The anchor costs at most 0.003 accuracy on public data. On synthetic flows it helps, reaching 0.960 accuracy ($p < 10^{-4}$ against both AE-SVM and RBF-SVM).

Kernel-target alignment [38] is lower for K_A than for the RBF kernel on every dataset (Table 3). This fits Corollary 1. The amplitude kernel has a small, fixed feature space and no width to tune.

5.2 Sample efficiency

Small-sample advantages are a common motivation for quantum kernels. We find none here (Fig. 2). On QKD data AE-SVM trails RBF-SVM by 0.03 to 0.04 MCC for $n = 50$ to 200 and catches up only at $n = 1000$. On synthetic flows it trails by 3.6 points at $n = 50$. On CIC-IDS2017 it trails at small n. On UNSW-NB15 it tracks RBF-SVM. The Laplacian kernel is the strongest kernel on public data.

The control tasks of Fig. 3 show that the bias can help when it matches the data. Inputs in $\mathbb{R}^8$ have heavy-tailed norms. When labels depend on direction only, AE-SVM learns from far fewer samples than any baseline. When labels depend on norm only, it stays at chance at every n, as Corollary 1 requires.

5.3 Unseen attack types

Table 4 and Fig. 4 hold the central result. On synthetic flows plain AE-SVM fails. Every held-out type scores between 0.45 and 0.61, against 0.839 on average for RBF-SVM.

QKD data first looks like a counter-example, since AE-SVM does well there. But one QKD feature, the Mahalanobis distance, is itself a norm. Removing it drops AE-SVM to 0.554. RBF-SVM barely changes. With the feature present, the encoding sees the norm through a coordinate, and the blind spot is covered.

The public benchmarks do not show the failure. AE-SVM matches or beats RBF-SVM on average. Figure 5 breaks this down by family. On CIC-IDS2017 most supervised models struggle on Bot and PortScan, and the benign-only autoencoder is the strongest detector. This matches the closed-world warning of Sommer and Paxson [9].

Table 2 In-distribution detection (mean ± 95% CI over seeds). QKD uses $s = 0.3$ and $n = 1400$. Bold marks the best mean per column. A dash means not run.

	QKD telemetry		Synthetic flows		UNSW-NB15		CIC-IDS2017	
Model	MCC	AUC	Accuracy	AUC	F1	AUC	F1	AUC
AE-SVM	.814±.023	**.970±.005**	.947±.002	.966±.003	.887±.003	.973±.001	.979±.001	.999±.000
AE-SVM (anchored)	.816±.019	.965±.007	.960±.003	.983±.002	.884±.003	.972±.001	.977±.001	.998±.000
RBF-SVM	.810±.015	.957±.007	.948±.003	.976±.003	.887±.003	.973±.001	.983±.001	.999±.000
Laplacian-SVM	–	–	.946±.003	.976±.002	.899±.004	.978±.001	.987±.001	**.999±.000**
Poly-SVM	–	–	.959±.003	.979±.003	.889±.003	.975±.001	.989±.001	.999±.000
GP	.801±.016	.956±.007	–	–	–	–	–	–
Hybrid	.833±.016	.969±.006	–	–	–	–	–	–
RF	.834±.013	.969±.006	.943±.003	.976±.002	.889±.003	.980±.001	.990±.002	**.999±.000**
RF-isotonic	**.837±.013**	.967±.006	–	–	–	–	–	–
XGBoost	–	–	.962±.003	**.985±.002**	.892±.003	**.981±.001**	**.994±.001**	**1.000±.000**
MLP	–	–	**.965±.001**	.984±.002	.889±.004	.975±.001	.991±.001	**.999±.000**
DNN	–	–	–	–	**.909±.006**	.977±.002	.989±.002	**.999±.000**
Autoencoder	–	–	–	–	.394±.187	.887±.009	.546±.058	.948±.012

Table 3 Kernel-target alignment, averaged over attack types.

Kernel	QKD	Synthetic	UNSW-NB15	CIC-IDS2017
AE	0.17	0.13	0.38	0.62
RBF	0.48	0.21	0.52	0.67

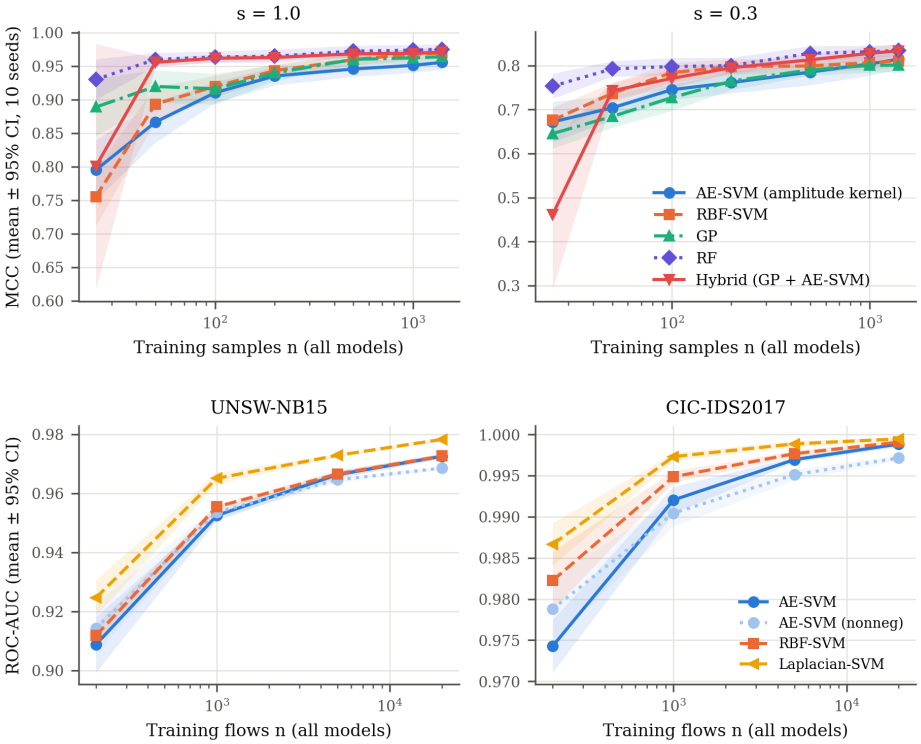


Fig. 2 Every model trained on the same subsample at each n (mean ± 95% CI). Top, QKD telemetry (MCC). Bottom, public data (ROC-AUC).

5.4 Mechanism

Corollary 1 offers two causes for the failure, sign ambiguity and norm invariance. Interventions separate them (Table 4). Scaling to $[0, 1]$ removes the sign ambiguity but keeps norm invariance. It helps little on synthetic flows (0.593). The anchor restores the norm. It closes most of the gap on synthetic flows (0.793) and all of it on QKD data without the Mahalanobis feature (0.939). On public data it changes little.

Table 5 varies c. The result follows the shape that Proposition 3 predicts. Small values leave the norm invisible. Large values crowd all states together. The median rule lands near the best value without seeing held-out attacks.

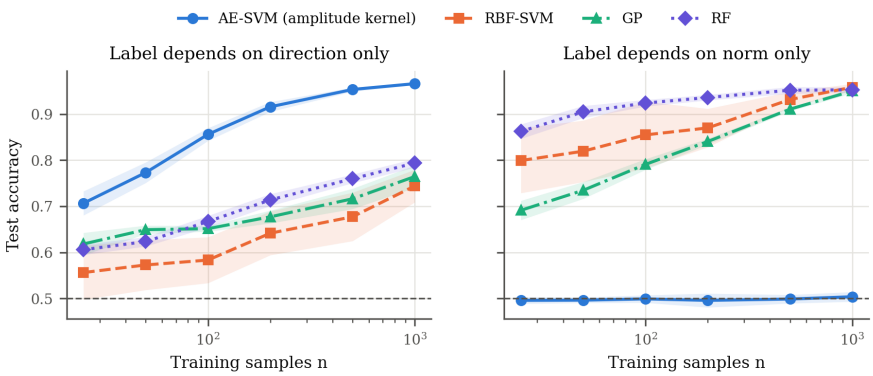


Fig. 3 Control tasks in $\mathbb{R}^8$ with heavy-tailed norms (mean ± 95% CI over 10 seeds).

Table 4 Mean ROC-AUC on held-out attack types. QKD uses $s = 0.3$. "No Mahal." removes the only QKD feature that carries magnitude. Bold marks the best per column. A dash means not run.

Model	QKD	QKD, no Mahal.	Synthetic	UNSW-NB15	CIC-IDS2017
AE-SVM	0.952	0.554	0.526	0.945	0.715
AE-SVM (nonneg)	–	–	0.593	0.935	0.663
AE-SVM (anchored)	0.949	0.939	0.793	0.940	0.733
RBF-SVM	0.917	**0.940**	**0.839**	0.932	0.658
GP	0.941	–	–	–	–
RF	**0.960**	–	–	0.949	0.820
XGBoost	–	–	–	0.960	0.852
DNN	–	–	–	**0.963**	0.694
Autoencoder	–	–	–	0.860	**0.874**

Table 5 Synthetic flows. Mean held-out AUC against the anchor constant c (5 seeds). $c = 0$ is plain encoding. The median rule gives $c \approx 6.4$.

c	0	1	2	4	8	16	32	median
AUC	0.526	0.549	0.691	0.809	0.771	0.650	0.521	0.793

5.5 An entangling feature map

Table 6 and Fig. 6 compare five kernels on the same eight inputs. The ZZ map is not norm invariant. It does not share the failure of plain AE on synthetic flows (held-out AUC 0.648 against 0.527). It is close to the classical kernels everywhere. But it is not the best kernel on held-out attacks in any of the six settings. The tuned Laplacian kernel is best in four of them.

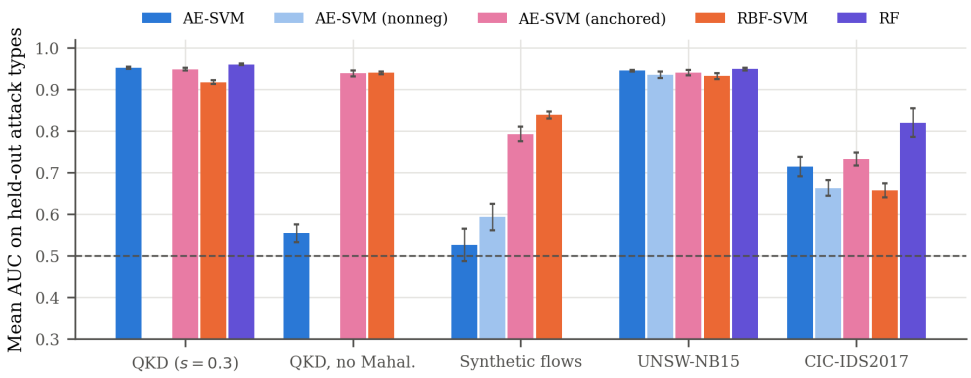


Fig. 4 Mean ROC-AUC on held-out attack types across domains (mean ± 95% CI over seeds).

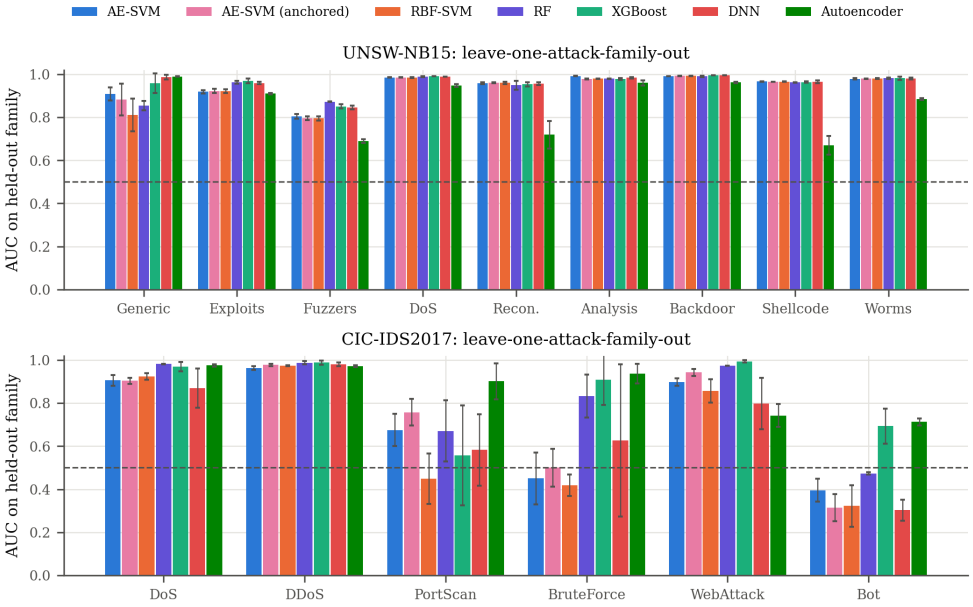


Fig. 5 Held-out AUC per attack family on the public benchmarks (mean ± 95% CI over seeds).

Reducing to eight inputs changes some results from the full-dimensional study. Held-out AUC on synthetic flows falls for every kernel. On the public data the anchor now lowers held-out AUC, by 0.036 on UNSW-NB15 and 0.116 on CIC-IDS2017. So the anchor is not free. It helps when the norm carries the signal and can hurt when it does not.

The centred QKD setting isolates the sign ambiguity. Standardizing the Mahalanobis column makes it signed. A block far from normal and a block unusually close

Table 6 Five kernels on the same eight-dimensional inputs, trained exactly on $n = 1000$ samples (mean ± 95% CI over 3 seeds, averaged over held-out types). Bold marks the best per row.

Domain	AE	AE (anchored)	ZZ	RBF	Laplacian
AUC on unseen samples of known attack types					
QKD	**0.974±0.002**	0.968±0.004	0.958±0.005	0.964±0.006	0.969±0.003
QKD, centred Mahal.	0.741±0.008	**0.972±0.003**	0.959±0.004	**0.972±0.003**	0.970±0.003
QKD, no Mahal.	0.730±0.012	**0.970±0.003**	0.950±0.003	0.969±0.002	0.968±0.004
Synthetic flows	0.949±0.006	**0.989±0.004**	0.983±0.003	0.988±0.004	**0.989±0.003**
UNSW-NB15	0.946±0.002	0.932±0.006	0.954±0.001	0.955±0.004	**0.964±0.003**
CIC-IDS2017	0.976±0.004	0.969±0.007	0.992±0.001	0.991±0.002	**0.993±0.001**
AUC on the held-out attack type					
QKD	0.945±0.003	**0.948±0.008**	0.941±0.009	0.927±0.015	0.942±0.005
QKD, centred Mahal.	0.568±0.027	0.929±0.012	0.916±0.013	0.932±0.009	**0.944±0.005**
QKD, no Mahal.	0.586±0.019	0.933±0.008	0.903±0.008	0.939±0.004	**0.940±0.008**
Synthetic flows	0.527±0.034	0.678±0.037	0.648±0.037	**0.690±0.030**	0.679±0.038
UNSW-NB15	0.820±0.010	0.783±0.009	0.821±0.009	0.821±0.002	**0.840±0.005**
CIC-IDS2017	0.744±0.123	0.628±0.114	0.874±0.065	0.848±0.070	**0.890±0.045**

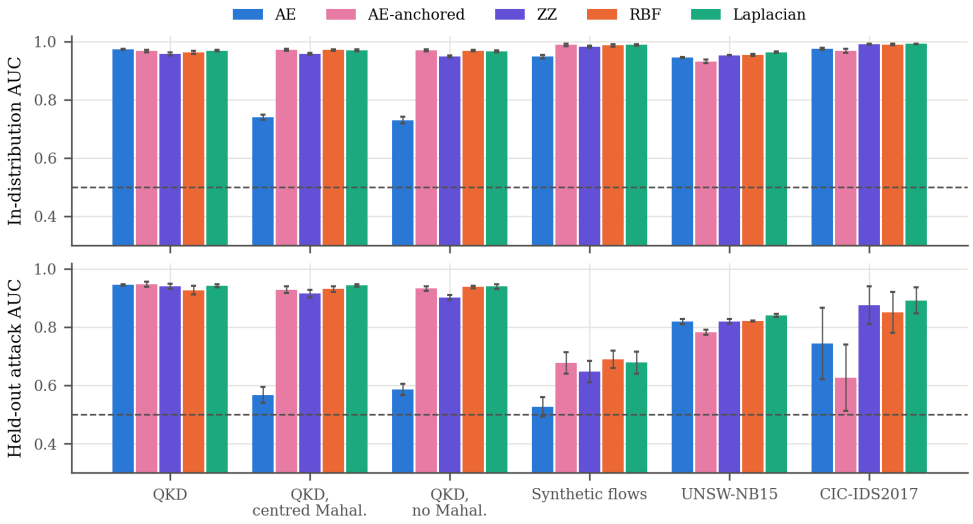


Fig. 6 Kernels on the same eight-dimensional inputs (mean ± 95% CI over seeds). The ZZ map uses eight qubits and a cross-validated bandwidth.

to normal then differ mainly in sign, which K_A cannot see. Plain AE falls from 0.945 to 0.568. The anchor restores 0.929.

Tuning the classical baseline mattered as much as the encoding. In a first run the RBF width grid stopped at $\gamma = 3/q$. On CIC-IDS2017 the ZZ map then appeared to beat RBF by 0.20 in held-out AUC (0.874 against 0.672). Extending the grid to $10/q$ raised RBF to 0.848 and the Laplacian kernel reached 0.890. The selected classical

Table 7 Off-diagonal ZZ kernel entries on training data (mean ± standard deviation). For eight qubits a fully concentrated kernel has mean $2^{-8} \approx 0.004$.

Domain	Selected β (median)	Entries at selected β	Entries at $\beta = 1$
QKD	0.02	0.641 ± 0.315	0.005 ± 0.009
QKD, centred Mahal.	0.05	0.464 ± 0.244	0.006 ± 0.010
QKD, no Mahal.	0.05	0.510 ± 0.235	0.011 ± 0.016
Synthetic flows	0.05	0.485 ± 0.261	0.005 ± 0.008
UNSW-NB15	0.2	0.091 ± 0.207	0.019 ± 0.089
CIC-IDS2017	0.2	0.036 ± 0.129	0.007 ± 0.034

Table 8 Change in mean AUC when every kernel entry is estimated from S shots, relative to exact kernels. Averaged over all six settings of Table 6. Training uses $n = 1000$, which needs about 5×10^5 entry estimates.

Kernel	Test	$S = 100$	$S = 1000$	$S = 10000$
AE-anchored	In-distribution	−0.008	+0.001	+0.001
	Held-out	−0.007	−0.002	−0.000
ZZ	In-distribution	−0.016	−0.004	−0.001
	Held-out	−0.016	−0.004	+0.001

widths on both public datasets still sit at the top of the extended grid. Wider grids might change the classical numbers again. We report the grid as run.

Bandwidth matters for the ZZ map as theory predicts (Table 7). At $\beta = 1$ the off-diagonal entries average 0.005 to 0.019, close to the fully concentrated value 2^{-8}. Cross-validation picks β between 0.01 and 0.8. There the entries spread out and the kernel can learn. This matches the concentration and bandwidth results of earlier work [18–20].

5.6 Finite shots

Shot noise costs little accuracy at the sample size we use (Table 8). With 1000 shots per entry, the worst setting loses 0.015 AUC for either kernel. With 100 shots the ZZ kernel loses up to 0.051 on QKD data, centred Mahal.. The anchored amplitude kernel loses at most 0.032. The real cost is the number of circuit runs. A training Gram matrix on 1000 samples at 1000 shots per entry needs about 5×10^8 executions, before any test point is scored. The closed form of Proposition 1 needs none.

5.7 Can the failure be predicted?

A practitioner would like to know in advance whether a dataset falls into the failure case. We tried the obvious statistic (Table 9). The norm alone separates attacks best on synthetic flows, where AE-SVM fails. But across held-out splits it does not predict the

Table 9 Norm-versus-direction diagnostic, averaged over held-out attack types. Norm AUC scores $\|\mathbf{x}\|$ alone. Direction AUC scores a direction-only distance. The last column is AE-SVM minus RBF-SVM held-out AUC.

Domain	Norm AUC	Direction AUC	Gap
QKD	0.76	0.58	+0.034
QKD, no Mahal.	0.76	0.59	−0.386
Synthetic flows	0.90	0.66	−0.313
UNSW-NB15	0.51	0.47	+0.013
CIC-IDS2017	0.64	0.67	+0.057

Table 10 QKD telemetry, $s = 0.3$. Deferral by uncertainty $u = \sqrt{p(1-p)}$. Coverage is the share of blocks kept.

	$u < 0.3$		$u < 0.1$	
Model	Coverage	Accuracy	Coverage	Accuracy
GP	69%	97.6%	0%	–
RF-isotonic	80%	97.6%	35%	98.8%
AE-SVM	79%	97.8%	31%	99.3%

gap ($\rho = -0.09$, $p = 0.65$, 28 splits). It also misses the QKD ablation. There, removing one coordinate barely changes the norm's separating power yet flips the outcome. Which features carry magnitude matters more than the norm's overall power. On real traffic, attack families also differ in which features are active, and that difference is visible to a direction-only kernel.

5.8 Calibration and deferral

Gaussian-process posteriors are often proposed as a source of uncertainty for security alerts. On QKD data the Laplace GP is the least calibrated supervised model, with expected calibration error [39] 0.047 ± 0.006. Isotonic RF is the best at 0.030. The GP's probabilities crowd the decision boundary (Table 10). It keeps fewer blocks than isotonic RF or AE-SVM at the same accuracy, and at $u < 0.1$ it keeps none.

6 Discussion

What the encoding decides. Amplitude encoding fixes a model class before any training. By Proposition 2 that class is the set of Rayleigh quotients. It contains good detectors only when the label is a function of direction. This one property accounts for the main pattern in our results. The bias helps on direction-only control tasks. It hurts when unseen attacks are mainly larger than normal traffic, as in both simulators. It

is harmless when attack families differ in which features are active, as on both public benchmarks. The same reasoning applies to any fidelity kernel whose state preparation normalizes its input. It also applies on hardware, since the invariance is a property of the state and not of how the overlap is estimated.

Simulators can mislead. Our simulators alone would have suggested that amplitude encoding is unsafe for zero-day detection in general. The public benchmarks reversed that. A simple diagnostic did not anticipate which case applied. We therefore recommend that quantum detectors be judged on held-out classes from real data, not on simulators alone.

Recommendations for quantum-kernel studies.

1. Train classical kernels on the same data at every sample size.
2. Derive what the encoding is invariant to before running experiments.
3. If amplitude encoding is used, test the anchored variant on held-out classes. The median anchor fixed every failure we saw. On full-dimensional data it cost at most 0.005 AUC elsewhere, but on eight-dimensional public data it cost more.
4. Tune the bandwidth of angle-encoded kernels and report how concentrated the kernel is. Tune classical baselines over grids wide enough that the chosen value is not at an edge.
5. Report held-out class results, and for simulated data report single-feature separability.

Limitations. QKD attack sizes are modeling assumptions, and to our knowledge no public QKD attack telemetry exists. The flow simulator is not packet-level traffic. Both public datasets have known labeling artifacts. We subsample them to 20,000 training flows, which favours kernel methods. Main experiment kernels were not tuned per dataset. The ZZ experiment uses PCA to eight inputs, which removes information that full-dimensional models can use. Its selected classical widths sat at the top of their grid on public data. We test one entangling feature map, at one depth and one qubit count. We simulate shot noise but not gate noise or readout error. Holding out one attack family only approximates a truly new attack.

7 Conclusion

Amplitude-encoding kernel machines compute Rayleigh quotients. They see the direction of an input and never its norm. At matched scale on four security datasets this kernel never beats a Gaussian kernel in distribution and needs more data when samples are few. In both simulated settings it fails on unseen attacks because it discards the norm. One anchoring amplitude, set to the median training norm, repairs this, and the theory explains why. On two public benchmarks the failure does not appear. An eight-qubit ZZ feature map with tuned bandwidth avoids the blind spot but did not beat tuned classical kernels. Encodings should be chosen and tested for their invariances, on real held-out classes, against classical kernels at the same scale.

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- **Ethics approval and consent to participate.** Not applicable.
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- **Data availability.** UNSW-NB15 and CIC-IDS2017 are publicly available from their original providers. The simulators that generate the QKD and synthetic flow data, and all result files, are at https://github.com/arnavk23/quasar-security.
- **Materials availability.** Not applicable.
- **Code availability.** All code is at https://github.com/arnavk23/quasar-security.
- **Author contributions.** A.K. designed the study, wrote the code, ran the experiments and wrote the manuscript.

References

[1] Havlíček, V., Córcoles, A.D., Temme, K., Harrow, A.W., Kandala, A., Chow, J.M., Gambetta, J.M.: Supervised learning with quantum-enhanced feature spaces. Nature **567**, 209–212 (2019) https://doi.org/10.1038/s41586-019-0980-2

[2] Schuld, M.: Supervised quantum machine learning models are kernel methods. arXiv:2101.11020 (2021)

[3] Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., McClean, J.R.: Power of data in quantum machine learning. Nat. Commun. **12**, 2631 (2021) https://doi.org/10.1038/s41467-021-22539-9

[4] Kübler, J.M., Buchholz, S., Schölkopf, B.: The inductive bias of quantum kernels. In: Advances in Neural Information Processing Systems, vol. 34 (2021)

[5] Liu, Y., Arunachalam, S., Temme, K.: A rigorous and robust quantum speed-up in supervised machine learning. Nat. Phys. **17**, 1013–1017 (2021) https://doi.org/10.1038/s41567-021-01287-z

[6] Bowles, J., Ahmed, S., Schuld, M.: Better than classical? The subtle art of benchmarking quantum machine learning models. arXiv:2403.07059 (2024)

[7] Kalinin, M., Krundyshev, V.: Security intrusion detection using quantum machine learning techniques. J. Comput. Virol. Hacking Tech. **19**(1), 125–136 (2023) https://doi.org/10.1007/s11416-022-00435-0

[8] Suryotrisongko, H., Musashi, Y.: Evaluating hybrid quantum-classical deep learning for cybersecurity botnet DGA detection. Procedia Comput. Sci. **197**, 223–229 (2022) https://doi.org/10.1016/j.procs.2021.12.135

[9] Sommer, R., Paxson, V.: Outside the closed world: On using machine learning for network intrusion detection. In: Proc. IEEE Symp. Security and Privacy, pp.

305–316 (2010). https://doi.org/10.1109/SP.2010.25

[10] Huttner, B., Imoto, N., Gisin, N., Mor, T.: Quantum cryptography with coherent states. Phys. Rev. A **51**(3), 1863–1869 (1995) https://doi.org/10.1103/PhysRevA.51.1863

[11] Lydersen, L., Wiechers, C., Wittmann, C., Elser, D., Skaar, J., Makarov, V.: Hacking commercial quantum cryptography systems by tailored bright illumination. Nat. Photonics **4**, 686–689 (2010) https://doi.org/10.1038/nphoton.2010.214

[12] Gisin, N., Fasel, S., Kraus, B., Zbinden, H., Ribordy, G.: Trojan-horse attacks on quantum-key-distribution systems. Phys. Rev. A **73**(2), 022320 (2006) https://doi.org/10.1103/PhysRevA.73.022320

[13] Bennett, C.H., Bessette, F., Brassard, G., Salvail, L., Smolin, J.: Experimental quantum cryptography. J. Cryptol. **5**(1), 3–28 (1992) https://doi.org/10.1007/BF00191318

[14] Moustafa, N., Slay, J.: UNSW-NB15: A comprehensive data set for network intrusion detection systems. In: Proc. Military Communications and Information Systems Conf. (MilCIS), pp. 1–6 (2015). https://doi.org/10.1109/MilCIS.2015.7348942

[15] Sharafaldin, I., Lashkari, A.H., Ghorbani, A.A.: Toward generating a new intrusion detection dataset and intrusion traffic characterization. In: Proc. ICISSP, pp. 108–116 (2018). https://doi.org/10.5220/0006639801080116

[16] Schuld, M., Sweke, R., Meyer, J.J.: Effect of data encoding on the expressive power of variational quantum-machine-learning models. Phys. Rev. A **103**, 032430 (2021) https://doi.org/10.1103/PhysRevA.103.032430

[17] LaRose, R., Coyle, B.: Robust data encodings for quantum classifiers. Phys. Rev. A **102**, 032420 (2020) https://doi.org/10.1103/PhysRevA.102.032420

[18] Thanasilp, S., Wang, S., Cerezo, M., Holmes, Z.: Exponential concentration in quantum kernel methods. Nat. Commun. **15**, 5200 (2024) https://doi.org/10.1038/s41467-024-49287-w

[19] Shaydulin, R., Wild, S.M.: Importance of kernel bandwidth in quantum machine learning. Phys. Rev. A **106**, 042407 (2022) https://doi.org/10.1103/PhysRevA.106.042407

[20] Canatar, A., Peters, E., Pehlevan, C., Wild, S.M., Shaydulin, R.: Bandwidth enables generalization in quantum kernel models. arXiv:2206.06686 (2022)

[21] Wang, X., Du, Y., Luo, Y., Tao, D.: Towards understanding the power of quantum kernels in the NISQ era. Quantum **5**, 531 (2021) https://doi.org/10.22331/

q-2021-08-30-531

[22] Hubregtsen, T., Wierichs, D., Gil-Fuster, E., Derks, P.-J.H.S., Faehrmann, P.K., Meyer, J.J.: Training quantum embedding kernels on near-term quantum computers. Phys. Rev. A **106**, 042431 (2022) https://doi.org/10.1103/PhysRevA.106.042431

[23] Tang, E.: A quantum-inspired classical algorithm for recommendation systems. In: Proc. 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC), pp. 217–228 (2019). https://doi.org/10.1145/3313276.3316310

[24] Silva, T.F., Xavier, G.B., Temporão, G.P., Weid, J.P.: Real-time monitoring of single-photon detectors against eavesdropping in quantum key distribution systems. Opt. Express **20**(17), 18911 (2012) https://doi.org/10.1364/OE.20.018911

[25] Lo, H.-K., Ma, X., Chen, K.: Decoy state quantum key distribution. Phys. Rev. Lett. **94**(23), 230504 (2005) https://doi.org/10.1103/PhysRevLett.94.230504

[26] Lo, H.-K., Curty, M., Qi, B.: Measurement-device-independent quantum key distribution. Phys. Rev. Lett. **108**(13), 130503 (2012) https://doi.org/10.1103/PhysRevLett.108.130503

[27] Mao, Y., Huang, W., Zhong, H., Wang, Y., Qin, H., Guo, Y., Huang, D.: Detecting quantum attacks: a machine learning based defense strategy for practical continuous-variable quantum key distribution. New J. Phys. **22**, 083073 (2020) https://doi.org/10.1088/1367-2630/aba8d4

[28] Buczak, A.L., Guven, E.: A survey of data mining and machine learning methods for cyber security intrusion detection. IEEE Commun. Surveys Tuts. **18**(2), 1153–1176 (2016) https://doi.org/10.1109/COMST.2015.2494502

[29] Mirsky, Y., Doitshman, T., Elovici, Y., Shabtai, A.: Kitsune: An ensemble of autoencoders for online network intrusion detection. In: Proc. NDSS (2018). https://doi.org/10.14722/ndss.2018.23204

[30] Möttönen, M., Vartiainen, J.J., Bergholm, V., Salomaa, M.M.: Transformation of quantum states using uniformly controlled rotations. Quantum Inf. Comput. **5**(6), 467–473 (2005) https://doi.org/10.26421/QIC5.6-5

[31] Bergholm, V., et al.: PennyLane: Automatic differentiation of hybrid quantum-classical computations. arXiv:1811.04968 (2018). https://doi.org/10.48550/arXiv.1811.04968

[32] Rousseeuw, P.J., Driessen, K.V.: A fast algorithm for the minimum covariance determinant estimator. Technometrics **41**(3), 212–223 (1999) https://doi.org/10.1080/00401706.1999.10485670

[33] Williams, C.K.I., Seeger, M.: Using the Nyström method to speed up kernel machines. In: Advances in Neural Information Processing Systems 13, pp. 682–688 (2001)

[34] Breiman, L.: Random forests. Mach. Learn. **45**(1), 5–32 (2001) https://doi.org/10.1023/A:1010933404324

[35] Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proc. ACM SIGKDD, pp. 785–794 (2016). https://doi.org/10.1145/2939672.2939785

[36] Rasmussen, C.E., Williams, C.K.I.: Gaussian Processes for Machine Learning. MIT Press, ??? (2006). https://doi.org/10.7551/mitpress/3206.001.0001

[37] Zadrozny, B., Elkan, C.: Transforming classifier scores into accurate multiclass probability estimates. In: Proc. ACM SIGKDD, pp. 694–699 (2002). https://doi.org/10.1145/775047.775151

[38] Cristianini, N., Shawe-Taylor, J., Elisseeff, A., Kandola, J.: On kernel-target alignment. In: Advances in Neural Information Processing Systems 14, pp. 367–373 (2002)

[39] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proc. ICML. PMLR, vol. 70, pp. 1321–1330 (2017)